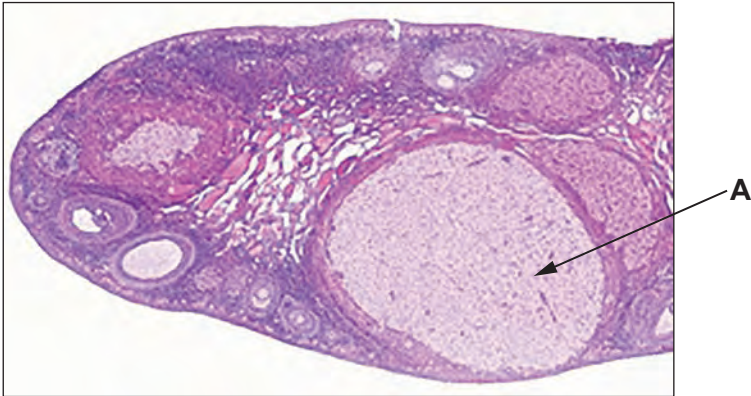


Answer **all** questions.

1. **Image 1.1** shows a section of human ovary.

Image 1.1



Following ovulation, structure **A** performs an endocrine function, secreting a hormone which reaches a maximum concentration about 6 days later. Active growth of blood vessels occurs in the ovary after ovulation.

(a) (i) Name structure **A** and state its function. [2]

.....

.....

(ii) Suggest **one** reason why it is important that the number of blood vessels in the ovary increase after ovulation. [1]

.....

.....

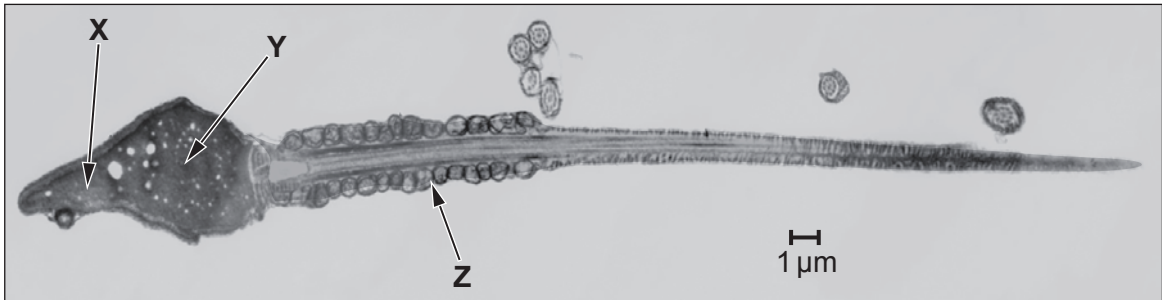
.....

.....



Image 1.2 shows a single human spermatozoon.

Image 1.2



- (b) (i) Calculate the actual length of the spermatozoon shown in **Image 1.2**. Give your answer in mm. [2]

Length = mm

- (ii) Structure **A** in **Image 1.1** has a mean diameter of 16 mm. Calculate the number of times larger structure **A** is than the human spermatozoon. [1]

Structure **A** is times larger than the human spermatozoon.

- (c) (i) Complete **Table 1.3** by identifying structures **X**, **Y** and **Z** on **Image 1.2**, describing their function during fertilisation. [3]

Table 1.3

Letter	Name of structure	Function during fertilisation
X		
Y		
Z		



- (ii) Humans normally have a diploid number $2n = 46$. Occasionally a secondary oocyte is fertilised by two sperm. The embryo formed develops abnormally and will not survive.
State the number of chromosomes this embryo would have and the term used to describe the number of sets of chromosomes. [2]

Number of chromosomes =

Term used

- (iii) This number of sets of chromosomes also occurs naturally in a certain plant tissue. State the name of this plant tissue and state its function. [2]

.....

.....

Image 1.4 shows a human placenta, following delivery, after the baby is born.

Image 1.4



- (d) Use **Image 1.4** to state **two** features of a placenta and describe how they are important to the survival of the embryo. [2]

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